Risk-Sensitive Reinforcement Learning for Joint Concentrated-Liquidity and Hedge Management on Uniswap v3

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Abstract—Concentrated liquidity provision on Uniswap v3 turns a passive role that only earns fees into an active control problem. A liquidity provider (LP) must decide when to recenter its price band, how wide to keep it, and how much of the position's price exposure to hedge, while fee income competes with impermanent loss, loss-versus-rebalancing (LVR), gas and slippage. Empirical studies show that fees frequently fail to cover these losses. Whether one learned policy can manage all three decisions jointly under realistic frictions, and where active management stops paying, remains open. We present Undertow, a research testbed with two engines: a replay backtester whose fee accounting stays in exact integer arithmetic across two years of on-chain Uniswap v3 events, and a fast Gymnasium simulator. We formulate liquidity provision as a risk-sensitive Markov decision process (MDP) with impulse control over a joint action space of timing, width and hedge ratio. Four claims structure the study: a negative limit that locates where fees fall short of LVR and rebalancing costs; a methods contribution documenting why naive PPO degenerates on this MDP and which fixes (band geometry in the observation, an excess-versus-HODL reward, environment diversity, and execution at decision time) are necessary; evidence from windows held out from training on whether a single risk-sensitive policy dominates the best static width and a hedged HODL benchmark matched in beta; and a map of the active/passive boundary in terms of volatility, fee tier, gas and capital. [PENDING: The abstract will close with the quantitative headline and the C3/C4 verdicts once the locked experiments across seeds land.]

Index Terms—reinforcement learning, Uniswap v3, concentrated liquidity, market making, loss-versus-rebalancing, impermanent loss, risk-sensitive reinforcement learning, simulation.

I. Introduction

AUTOMATED market makers (AMMs) turned market making into a public utility. A liquidity provider (LP) deposits a pair of assets into a smart contract, and the contract prices every trade through a deterministic trading function. Uniswap v3 refined that bargain with concentrated liquidity: an LP chooses a price interval, and the deposited capital backs trades only inside it [1]. Because fee accrual is proportional to the share of active liquidity, narrowing the interval raises the fee earned per dollar of capital while the price stays inside the band. This is the promise that drew capital to v3: the same deposit can quote with ten, a hundred, or a thousand times the fee density of a full-range position.

The catch is that liquidity quoted at a stale price is an open invitation to traders with better information. Every arbitrage trade that moves the pool price toward the market price is a trade against the LP, and the resulting loss does not reverse when the price moves back. Milionis et al. formalized this cost of adverse selection as loss-versus-rebalancing (LVR): the shortfall of a constant-product LP position relative to a hypothetical portfolio that continuously rebalances to the same token mix [2]. For a full-range constant-product pool, LVR accrues at one eighth of the square of volatility times the position value, a rate that grows with the very activity that generates fees. The empirical record is unfavorable. Fritsch and Canidio compare the arbitrage losses large AMM pools actually realized with the fee income those pools produced: in their sample a passive full-range deposit would have outperformed the concentrated alternatives [3]. Loesch et al. reach a related conclusion from v3 positions directly: concentration magnifies impermanent loss when the price leaves the band, so a narrow range can underperform even a v2 deposit [4]. Managing a range is not free either: hitting the edges forces a recentring trade that pays pool fees plus slippage, and each action costs gas.

The LP therefore faces a sequential decision problem, not a portfolio choice. At every decision instant it must decide whether to act, where to place the next band and how wide to make it, and how much of the position's directional exposure to hedge. The levers interact: a wider band earns less per trade but defers the next costly recentring; a hedge removes directional variance but pays funding and perp fees; acting sooner protects fee income but multiplies transaction costs. The heuristics in common use (recenter when the price exits the band, rebalance when the expected fee gain covers gas, hold a wide static band [1]) are single points in this space, chosen by argument rather than learned from outcomes.

Reinforcement learning (RL) is a natural fit for the joint problem, and recent work has begun to explore it. Zhang et al. train a hedged double deep Q-network agent for v3

liquidity provision and report sizeable gains over baselines, but their hedge executes without cost or funding, and the policy sees no regime features [5]. ZeroSwap learns to set the spread of a market maker so that it tracks an external price, an orthogonal lever to range placement, and targets zero profit against competitive counterparties [6]. RAmmStein derives rebalancing thresholds that depend on the regime through impulse control and reports that a policy which rebalances lazily when the regime warrants it improves net ROI while cutting rebalance counts by roughly 85 %, but its net metric excludes impermanent loss, it studies a single pair, and it relies on off-chain market data [7]. Chionas et al. study concentrated liquidity as a problem of risk aversion in simulation and find that more volatile and more expensive environments push optimal positions wider and less frequently rebalanced, with aversion to risk compressing the left tail of returns; their analysis, however, is simulation only, uses no on-chain data, and leaves the hedge leg open [8]. What is missing is a study that (i) evaluates policies on on-chain replay, (ii) charges every friction (gas, swap fees, slippage, perp costs and funding) on both the LP and hedge legs, (iii) optimizes and reports Sharpe and CVaR rather than raw profit, and (iv) treats rebalance timing, band width and hedge ratio as one joint action.

This paper builds that study around four claims.

- C1 Limits. On the pools and windows we study, passive fee income alone does not cover the sum of LVR, impermanent loss and rebalancing costs; we quantify the decomposition and the cost of greedy rebalancing, extending the LVR literature with an exact on-chain replay engine.
- C2 Failure modes. Naive RL fails on this Markov decision process (MDP). We document degenerate policies that collapse to a single action, identify what breaks (an observation in which the live band is hidden, a reward whose benchmark drifts away from the reported metric after a rebalance, and training windows that never change), and show that each fix (band geometry in the observation, an excess-versus-HODL reward, and window resampling) is necessary.
- C3 A learned joint policy. A risk-sensitive policy that controls band timing, band width and hedge ratio jointly beats the strongest family of rules tuned by hand on data held out from training, and the way it splits on each feature recovers the underlying economics: band exit, the trade-off between fees and gas, and volatility regime.
- C4 The active/passive boundary. We map the region in which active management beats a passive or hedged HODL benchmark as a function of volatility, fee tier, volume relative to liquidity, gas price and capital.

The claims are ordered deliberately: C1 establishes the economic terrain, C2 explains why the obvious RL approach fails there, and C3–C4 show what a policy that survives the terrain can do. To measure all four against ground truth, we built Undertow, a testbed that couples an exact replay backtester over two years of on-chain Uniswap v3 events, with fee accounting kept in integer arithmetic, to a fast float64 Gymnasium simulator, and we place the two engines under a parity harness so that the simulator can be trusted where it is cheap and checked where it matters [9]. Section II establishes the mechanics and the economics of concentrated liquidity. Section III positions the work against the literature. Section IV states the risk-sensitive MDP with impulse control. Section V describes the Undertow testbed and the failure modes it was built to survive. Section VI fixes the experimental protocol, Section VII reports C1–C4, and Section VIII discusses validity and the gap between simulation and reality. Section IX concludes.

II. Background: Mechanical and Economic Setting

The pair is USDC/WETH: token0 is USDC and token1 is WETH, as in the on-chain pools we use. Protocol quantities are raw Uniswap quantities: s denotes the raw square root of the pool price ($\sqrt{P}$, increasing in tick) and L the raw liquidity. The ETH price in USD is

$$q(s) \;=\; \frac{10^{\,d_1-d_0}}{s^2}, \tag{1}$$

roughly 3000 USDC/WETH at $s \approx 1.8 \times 10^4$, with token decimals $d_0 = 6$ and $d_1 = 18$; because q decreases in s, a lower tick means a higher ETH price. Monetary quantities are always converted to USDC, and the decimal factors cancel from every ratio we report.

A. Concentrated liquidity in Uniswap v3

Uniswap v3 replaces a single constant-product curve with a family of curves indexed by integer ticks [1]. An LP selects a tick interval $[t_\ell, t_u]$, equivalently a raw band $[s_a, s_b]$ with $s_a < s_b$, and deposits liquidity L that backs trades only while the pool price stays inside the band. For $s \in [s_a, s_b]$, the position holds

$$\begin{aligned} x(s) &= \frac{L}{10^{d_0}}\left(\frac{1}{s} - \frac{1}{s_b}\right), \\ y(s) &= \frac{L}{10^{d_1}}(s - s_a)\,, \end{aligned} \tag{2}$$

in human token units, so its USDC value is

$$V(s) \;=\; x(s) + y(s)\,q(s) \;=\; \frac{L}{10^{d_0}}\left(\frac{2}{s} - \frac{1}{s_b} - \frac{s_a}{s^2}\right). \tag{3}$$

Equation (3) makes the payoff structure visible. As the ETH price falls (s rises) toward the band, the position rotates into WETH; as the price rises (s falls), it rotates into USDC. At $s = s_b$ it holds only WETH, and above the band it is a pure WETH holding whose USDC value falls as the ETH price falls; symmetrically, at $s = s_a$ it holds only USDC. A v3 position therefore has a truncated, concave payoff in the price, in effect a short straddle around the current tick: it sells the asset that is appreciating and buys

the one that is depreciating, and it stops doing so once the price leaves the band.

Fee income accrues pro rata to the liquidity a position contributes at each tick crossed. The pool records cumulative fee growth per unit of liquidity, so a position that is in range over an interval earns $L \cdot \Delta\phi$ and a position that is out of range earns nothing [1]. A narrow band raises the LP's share of active liquidity for a given deposit, which multiplies fee density, but it shortens the expected time until the price exits and hence increases the frequency of costly recentring. This tension between fee density and band lifetime is the heart of the control problem.

B. Impermanent loss and loss-versus-rebalancing

A passive LP's return is not the pool's fee yield; it is fee income net of the change in the position's value relative to simply holding the initial deposit. For a full-range constant-product position initially split evenly in value, if the ETH price ratio is $r = q_t/q_0$, the position is worth $\rho(r) = 2\sqrt{r}/(1+r)$ relative to the initial basket, so the impermanent loss (IL) is

$$\mathrm{IL}(r) \;=\; \frac{2\sqrt{r}}{1+r} - 1 \;=\; -\frac{1}{8}(r-1)^2 + O\big((r-1)^3\big)\,, \quad (4)$$

which is second order in the price move; for example, $\mathrm{IL}(1.2) = -0.41\,\%$. On v3 the same formula applies only inside the band; outside it the position holds only one of the two assets and the comparison depends on where the band was placed [4].

IL measures the loss relative to the LP's own initial mix, but there is a sharper benchmark. Suppose a portfolio continuously rebalances to hold the same instantaneous token0 exposure as the LP position. Expanding the full-range value function $V(P) = 2\sqrt{kP}$ (in token1 units, with constant product k) to second order shows that the LP underperforms that portfolio, which replicates the LP's delta, at the rate

$$\mathrm{dLVR}_t \;=\; \frac{1}{8}\,\sigma^2\,V_t\,\mathrm{d}t, \quad (5)$$

which is the loss-versus-rebalancing of Milionis et al. [2]. Adverse selection therefore compounds with the square of volatility and linearly with the value at risk, and fee income must exceed it for passive provision to be viable. Concentrating liquidity raises the fee density but not the fee rate, so it improves the LP's position only when the extra fee share outweighs the extra exposure to arbitrage. The empirical studies in Section III find that, on many large pools, it does not.

C. The liquidity provider's control problem

The quantities above describe a static position. In practice an LP manages a sequence of decisions. Each recentring is an impulse: it incurs a fixed gas cost, a pool fee on the inventory imbalance that must be swapped to build the new range, and slippage. The impulse resets the band's geometry, and with it the fee density and the LVR exposure. The LP's choices are therefore (i) when to recenter, (ii) where and how wide to place the next band, and (iii) how much market risk to carry, either by holding the endogenous direction of the position or by hedging it with a short perpetual futures position whose notional tracks the LP's delta.

Two features make this a problem of impulse control rather than a static optimization [7]. First, the cost of acting is lumpy: a rebalance is worth doing only when the expected improvement in fee income, band lifetime or risk exceeds a threshold. Second, the state that determines whether to act is only partially observed: the volatility regime is latent, order flow is adversarial, and the LP sees history through delayed, aggregated on-chain events. Risk preferences enter as much as expected return does. An LP that maximizes expected fees will accept positions whose return distribution has a heavy left tail; an LP averse to risk will trade expected return for a compressed tail. We therefore evaluate policies by annualized Sharpe ratio and by conditional value at risk at the 5 % level (CVaR), a coherent risk measure expressible as a convex optimization [10], and we report raw PnL and excess-versus-HODL alongside them. Section IV turns this description into a formal MDP.

III. Related Work

A. The economics of passive liquidity

The modern account of AMM LP returns begins with adverse selection. Milionis et al. characterise the loss a constant-product LP suffers relative to a continuously rebalanced portfolio, name it loss-versus-rebalancing, and derive the local rate $\frac{1}{8}\sigma^2 V$ for a full-range position; the loss does not require informed traders to be modeled explicitly, since any price path generates it, and protocol fees and the choice of curve shift the balance [2]. Their framework implies a viability condition that we test against recorded pool histories in C1: fee income must exceed the LVR rate.

Whether that condition holds in practice has been tested on-chain. Fritsch and Canidio measure realized arbitrage losses against fee income across many large AMM pools and find that the losses typically dominate; in their sample a passive full-range position would have fared better than concentrated alternatives, and even a v2 deposit would have outperformed v3 in several pools [3]. Loesch et al. study v3 positions directly and show that concentrating liquidity magnifies impermanent loss when the price leaves the band: a narrow position can lose more than a v2 deposit over the same path [4]. Undertow replays each pool's own events, so the fee and impermanent loss terms in C1 come from the accumulators the protocol wrote.

B. Reinforcement learning for liquidity provision

Several recent papers apply RL to the LP problem, each isolating a different lever. Zhang et al. train a hedged double deep Q-network agent and report improvements of

9–69 % over baselines, but their hedge is assumed frictionless (their Assumptions 2–3), the policy receives no regime features, and the evaluation spans four test periods; they also document that the unhedged agent's PnL changes sign with the market trend, which is precisely the directional risk a hedge is meant to remove [5]. ZeroSwap takes an orthogonal route: instead of choosing where to place liquidity, it learns the spread of a market maker so that quoted prices track an external reference. In their competitive regime with zero profit the learned spread is the quantity that matters, and a static constant-product curve converges to the reference considerably more slowly [6]. RAmmStein derives rebalancing thresholds that depend on the regime as a problem of impulse control: greedy rebalancing loses 8.4 % in their setting, while a policy that rebalances lazily when the regime warrants it gains 1.6 % net ROI with roughly 85 % fewer rebalances. Their net metric excludes impermanent loss, the study covers a single pair, and it relies on off-chain market data [7]. Chionas et al. study concentrated liquidity as a problem of risk aversion in simulation: higher volatility and higher gas push optimal positions wider and reduce rebalancing, and an objective that penalizes risk trims the adverse tail of the return distribution. Their analysis is simulation only, uses no on-chain data, and does not consider a hedge leg [8].

A separate line of work learns market making quotes in the classical limit order book, where the spread and inventory are the control variables and the objective is typically a trade-off between mean and variance over inventory risk [11]. That formulation supplies the risk-sensitive vocabulary we use, but its state is a lit order book and its actions are quotes, not range geometry.

C. Tools and evaluation

We use Proximal Policy Optimization (PPO) with clipping as the learning algorithm [12], implemented through Stable-Baselines3 on Gymnasium environments [9], [13]. For evaluation under risk we use CVaR at the 5 % level, a coherent risk measure optimizable through its Rockafellar–Uryasev convex representation [10], alongside the annualized Sharpe ratio.

D. Positioning

Across this literature the pieces exist separately: the economics of LVR, the idea of learning a rebalance policy, the idea of a hedge, and the idea of risk aversion. A single study that trains over the joint action of rebalance timing, band width and hedge ratio, charges every friction on both legs, and evaluates on validated on-chain replay under an explicitly risk-sensitive metric has not been established. Undertow is built to close that gap. Its exact backtester makes the fee and LVR accounting a measured quantity rather than a modeling assumption, its float64 simulator makes the joint action space trainable, and the parity harness between them bounds the gap from simulation to reality that earlier studies which only simulate leave open [8]. The rest of this paper formalizes the MDP (Section IV), describes the testbed (Section V), and reports the four claims of Section I in order (Section VII).

IV. Problem Formulation

We formulate liquidity provision as a risk-sensitive Markov decision process with impulse control. The LP observes a feature vector built from past data, chooses between holding and executing an action that rebalances and hedges, and receives a reward at each step measuring the change in its excess return over a passive benchmark. The process is partially observed: the full market state is never available, so we work with the observation map defined below.

A. Episodes, state and observation

Time is discrete at the pinned decision cadence of $\Delta = 10\,\text{min}$. An episode covers a contiguous window of $H = 30$ days drawn from the training split, giving $T = 4320$ decision steps; the window's starting date is resampled in every training episode (Section V) so that the policy cannot memorise a single market path.

The environment's hidden state s_t comprises the market history up to t (reference prices, realized volatility, volume, active liquidity, gas), the LP's live position (raw band, liquidity, uncollected fees), and the hedge book (short notional, funding accrued). The agent sees the observation $o_t \in \mathbb{R}^{24}$, a vector with a fixed order in three groups:

- Market (indices 0–5, 11–15): human reference price, raw $\sqrt{P}$, tick, 1 h and 24 h log returns, annualized 24 h realized volatility, pool active liquidity, pool fee tier, recent gas price, recent ETH reference price, and the regime label (ordinal).
- Position (indices 6–10, 18–20): whether the position is in range, position value, uncollected fees in token0 and token1, impermanent loss against the fixed initial HODL basket, and the band geometry, meaning the signed distances from the current tick to the lower and upper bounds and the band width, all in tick-spacing units and positive while the position is in range.
- Hedge and time (indices 16–17, 21–23): short size as a fraction of the LP's ETH delta, hedge equity as a fraction of initial capital, the most recent 8 h funding rate, and the step index and steps remaining.

Every feature is closed on the right at the decision time: rolling windows use only bars at or before t, and the reference price is the last close no later than t. The features with heavy tails (price, tick, position value, fees, liquidity, gas and ETH price) are compressed with a signed log to $O(1)$ before they reach the network, a strictly monotone transform that preserves ordering and signs (ADR-034).

The band geometry group deserves emphasis. The action space below is written in tick-spacing units relative to the current tick, and the baselines are triggered by distance to the band edges. Without those three features

a memoryless policy cannot tell how close it is to a band exit from price and impermanent loss alone, which was the first failure mode we encountered (Section V).

B. Action space

At each decision epoch the agent selects the factored discrete action

$$a_t = (\kappa_t,\, c_t,\, w_t,\, h_t), \tag{6}$$

where

- $\kappa_t \in \{\text{hold}, \text{rebalance}\}$ is the impulse decision;
- $c_t \in \{-4, \ldots, +4\}$ shifts the center of the new band in tick-spacings around the current tick (used only when $\kappa_t = \text{rebalance}$);
- $w_t \in \{1, 2, 5, 10, 25, 50, 100, 250, 500, 1000\}$ is the band half width of the band in tick-spacings, snapped to the pool's spacing;
- $h_t \in \{0, 0.25, 0.5, 0.75, 1.0\}$ is the target fraction of the LP's ETH delta that the short perpetual hedge carries (0 = unhedged; 1 = the short carries the full LP delta).

The LP head alone has $1 + 9 \times 10 = 91$ options before hedging. The factored encoding keeps the action space small enough for a modest PPO setup while covering the three decisions that the economics of Section II identify. The current testbed implements the deterministic special case $h_t \equiv \bar{h}$ with $\bar{h}$ set in configuration; the discrete hedge ratio head is the joint policy extension evaluated for C3.

C. Reward: the change in excess return

A rebalance converts inventory between token0 and token1 and pays for the privilege. Writing the position value as V_t, the fixed initial HODL basket as H_t, and the cash flows from the hedge at each step as π_t^H (mark PnL), ϕ_t (funding) and τ_t^H (signed perp trading cost), the learning reward is

$$\begin{aligned} r_t = \frac{1}{W_0}\Big(&\text{fees}_t - \text{gas}_t - \text{slip}_t \\ &+ \Delta(V_t - H_t) + \pi_t^H + \phi_t + \tau_t^H - \lambda\, \hat{\sigma}_t^r\Big), \end{aligned} \tag{7}$$

where fees are nonnegative income, gas and slip enter as nonnegative cost magnitudes, $\Delta(V_t - H_t)$ is the signed change in the strategy's excess over the passive basket, W_0 is the initial capital, $\hat{\sigma}_t^r$ is a trailing estimate of reward volatility, and $\lambda \geq 0$ is a dial for risk aversion ($\lambda = 0$ by default). The imperative is excess versus a fixed initial basket: H_t is the value of the episode's opening deposit held passively, never rebased at a rebalance. This makes $\Delta(V_t - H_t)$ the change, from one decision to the next, in how far the strategy's value exceeds the passive basket, and makes the episode sum equal to the quantity the evaluation reports. The hedge leg is included with its true signs: a positive funding rate pays the short, while trading the perp costs $\tau_t^H \leq 0$. Dividing by W_0 keeps value targets $O(1)$ without hiding any cost.

The reward decomposition in Eq. (7) already exposes the economics of C1: fees are the only positive LP term, and LVR enters implicitly through $\Delta(V_t - H_t)$ as the adverse selection component of the excess change. The evaluation ledger records every term separately so that the decomposition can be reported directly.

D. Risk-sensitive objective and evaluation metrics

The learning objective is the usual discounted return with a coherent risk adjustment on the loss distribution $L = -G$, where G is the episode's total reward. Using the conditional value at risk at level α,

$$\text{CVaR}_\alpha(L) = \min_{\tau \in \mathbb{R}} \left\{ \tau + \frac{1}{1-\alpha}\, \mathbb{E}\big[(L - \tau)^+\big] \right\}, \tag{8}$$

the policy is trained to maximize $\mathbb{E}[G] - \lambda\, \text{CVaR}_\alpha(L)$ [10]. In the experiments of Section VII we report the two primary metrics directly: the annualized Sharpe ratio of per-episode returns and CVaR at the 5 % level (so $\alpha = 0.95$ in Eq. (8)), with raw PnL and excess-versus-HODL as secondary statistics. All four are computed from the recorded equity curve of each episode and aggregated across seeds with confidence intervals.

E. Hedge leg and benchmarks

The hedge is a short perpetual position margined in USDT and sized as a fraction h_t of the LP's ETH delta, $Q_t^H = h_t \cdot \partial V_t / \partial q_t$, where the LP delta is computed at the current band. Over one step its mark PnL is $\pi_t^H = -Q_t^H (q_{t+1} - q_t)$ plus the funding accrual ϕ_t at the prevailing 8 h rate, and trading to a new target pays a taker fee. Three reference points make the hedge result interpretable: the unhedged LP, the deterministic overlay $h_t \equiv \bar{h}$, and a hedged HODL portfolio matched in beta that holds the initial basket and shorts the same notional. The last is the benchmark that prevents us from claiming success against a weaker opponent: a policy that merely removes directional risk must still beat a passive portfolio with the same risk profile.

V. The Undertow Testbed

Answering C1–C4 requires an instrument, not an argument. Undertow is that instrument: a testbed with two engines in which the same policy can be trained cheaply and then measured against exact accounting on real on-chain events. Figure 1 sketches the data flow; this section describes each stage and, just as importantly, the failure modes the testbed was built to survive.

A. Data: an on-chain event tape

The data layer ingests the raw logs of two real USDC/WETH pools, the 0.05 % primary study pool (0x88e6a0…f5640, tick spacing 10) and the 0.30 % contrast pool (0x8ad599…e6d8, spacing 60), and converts them into a single event tape ordered by block and containing swaps, mints, burns and collect events. Alongside the tape it builds three auxiliary feeds: reference price closes

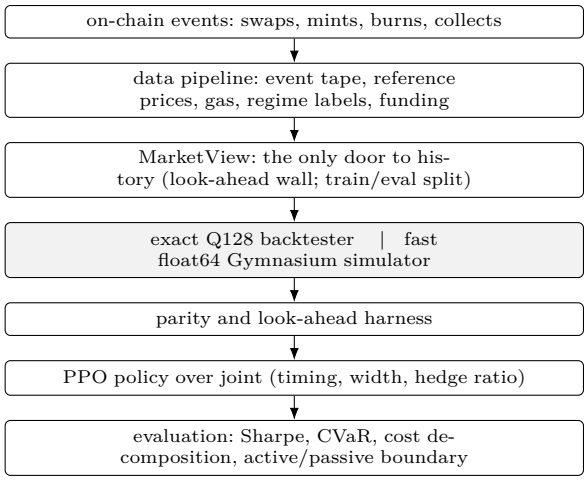


Fig. 1. Undertow: from on-chain events to a risk-sensitive policy. The two engines share one MarketView and are reconciled by the parity harness before any training result is reported.

at a resolution of one minute, gas prices at each block, and regime labels produced by thresholds committed in advance in the data module. A fourth feed, funding, is pulled from the Binance perpetual market margined in USDT at its native 8 h cadence and used by the hedge leg. The pinned window is 2022-01-01 to 2024-12-31 UTC; the walk-forward split trains on 2022–2023 and evaluates on 2024, so the test year is never seen during training or calibration. The primary pool's training lane alone carries roughly 4.86×10^6 events, which is why the throughput work described below is a necessity rather than an optimization.

Two design rules protect the results. First, the tape is the ground truth: prices, liquidity events and fees are the recorded on-chain values, not a fitted process. Second, all access goes through MarketView, which enforces both the block ordering and the train/eval boundary, so an observation at time t cannot read a bar from $t+1$ even by accident.

B. The exact backtester

The backtester replays a frozen policy against the tape using integer arithmetic. Fee accrual is computed in Q128 fixed point from the pool's recorded fee growth accumulators, exactly as the protocol does; liquidity events update an external tick lattice seeded from a boundary snapshot, and swap crossings are apportioned piecewise across the ticks they cross. The resulting ledger satisfies an accounting identity by construction,

$$\text{net} = \text{fees} + \text{il} - \text{gas} - \text{slippage} = \text{excess_vs_hodl}, \quad (9)$$

with impermanent loss measured against the fixed initial deposit basket. The policy protocol exposes no learning method, since a backtested policy is frozen by construction, and every ledger records the config hash, seed and git commit that produced it.

C. The fast simulator

Training on the exact engine is infeasible at millions of steps, so the simulator mirrors the same economics in float64. Its agent is marginal: agent liquidity never moves the price path, and the fee share is $\ell_{\text{agent}}/(\ell_{\text{agent}} + \ell_{\text{pool}})$ evaluated against the tape's recorded active liquidity. The tick lattice supports both minted agent positions and the external pool lattice, and the bookkeeping of snapshots and the tape path uses exact Python integers so that a large gross liquidity residue cannot silently corrupt a small delta, while a burn larger than the position still raises. Two price modes share one interface: replay, which walks the reference closes at a resolution of one minute, and a jump diffusion whose latent regimes follow a Markov chain with four states anchored to the regime labels, with drift and volatility for each state, Student-t jumps, and transitions estimated from the training split only. A throughput pass replaced scans at each step with indexes computed as of that step, cached the lattice keys and window views, and moved the vectorised environments into separate processes; on the pinned hardware the PPO throughput of the full pipeline rose from roughly 150–175 to about 450 steps per second. Actions execute at the decision price observed before the interval advances, matching the backtester and the latency a live LP would face.

D. Parity and look-ahead

Two numeric worlds are only useful if their disagreement is bounded. The parity harness feeds the simulator and the backtester the same price path, positions and gas series and compares fees, impermanent loss and PnL under a stated relative tolerance (10^{-3} for float64 against exact integers). After the fee engine was made exact, the remaining drift on the replay of real data is small: mean fee differences within each seed are on the order of 10^{-7} of notional, and impermanent loss matches exactly, comfortably inside tolerance. The harness originally exercised only the HOLD policy, which never rebalances; it now also runs a policy that rebalances, which is precisely the path the C1–C3 fees depend on. Alongside parity, a suite of look-ahead probes attacks the composed system: each component is checked to raise rather than read the future, and the probes include deliberately adversarial windows at the train/eval seam.

E. Learning and the failure modes it survived

The learning setup is deliberately modest: PPO with clipping on a 2×256 MLP, discount 0.99, generalized advantage estimation (GAE) with $\lambda = 0.95$, clip 0.15, five fixed seeds, and one million timesteps per seed. That simplicity is a feature: if a plain PPO cannot learn the task, the task or the interface is broken, and it was. The first two runs collapsed instead. Each seed locked onto one deterministic action, some always holding and some always repeating a single rebalance, and the measured advantage

did not separate the actions, with a correlation across actions near zero. Diagnosing that collapse produced three interface fixes that are now part of the testbed and are ablated for C2. The observation gained the three band geometry features, and the features with heavy tails gained signed log compression. The reward term for impermanent loss was moved from a rolling entry basket to the fixed initial HODL basket so that the reward at each step equals the change in the reported excess metric; the old parity harness could not catch that divergence because it only ran HOLD. Episodes gained window resampling: a reset without an episode seed now advances to a new window of 30 days instead of replaying the same lane for a million steps. Two cost defects were fixed on the way: rebalance slippage is charged on the inventory imbalance actually swapped rather than the whole position value, and the marginal fee credit is capped at the augmented denominator share so that a tiny agent cannot claim an outsized fee fraction. Each fix ships with a unit test and a decision record; the C2 table in Section VII lists symptom, diagnosis, fix and evidence.

F. Evaluation and reproducibility

The evaluation runner freezes a checkpoint and scores it on windows held out from training, emitting each episode's equity curve, cost ledgers and regime slices, then aggregating annualized Sharpe, raw PnL and excess-versus-HODL with confidence intervals across seeds and regimes. [PENDING: CVaR at 5 % exists as a metric in the metrics layer; wiring it into the aggregated evaluation report is the S21 deliverable, after which it joins the primary statistics.] The RQ1 ablation table, the RQ2 regime matrix and the report on the gap between simulator and backtester are generated from the same runner, and a dashboard that is read only replays policies against price and ranks runs by Sharpe out of sample. Every artefact carries the config hash and commit, so a reported number can be traced to the exact run that produced it.

VI. Experimental Design

The claims of Section I are only as credible as the protocol that tests them, so the protocol is fixed before any result is inspected, following the binding measurement protocol of ADR-030.

A. Data, windows and episodes

The study uses two real USDC/WETH pools: the 0.05 % primary pool (spacing 10) and the 0.30 % contrast pool (spacing 60). The study period runs from 2022-01-01 to 2024-12-31 (UTC) and is divided walk-forward: 2022–2023 supplies the training data and 2024 is held out for evaluation. Training episodes are windows of 30 days sampled from the training split with the environment's window resampling; evaluation tiles the 2024 split into nonoverlapping windows of 30 days (thirteen windows, the last partial), and no episode crosses the split boundary.

TABLE I
Benchmark policies. Each row is a frozen rule; the learned policy is compared against all of them, and C3 is claimed only if it beats the best static width and the hedged HODL matched in beta, not just hodl.

Benchmark	Rule
hodl	Hold the initial token mix; never provide liquidity.
passive_narrow	Deploy a ±5 % band once; never recenter.
passive_wide	Deploy a ±20 % band once; never recenter.
full_range_v2	Full range, equivalent to Uniswap v2.
tau_reset	Recenter whenever the price exits the live band.
cost_aware	Recenter at most daily, and only when the estimated fee gain exceeds gas.
Best static width	The ladder width with the best Sharpe on the training split, frozen for test.
Hedged HODL	Initial basket plus a short perp matched to its beta, with funding and perp costs.
Learned policy	PPO over the joint (timing, width, hedge ratio) action space.

Each episode runs at a decision cadence of 10 minutes with 100 000 USDC of agent capital, which keeps the agent marginal relative to pool liquidity and the fee share assumption honest. Every configuration in the study is trained with the five fixed seeds $\{0, 1, 2, 3, 4\}$ and one million timesteps per seed.

B. Benchmarks

A learned policy that beats only the weakest rule written by hand has demonstrated nothing. Our benchmark ladder therefore includes the full family of six policies, two additional reference points, and the hedged HODL benchmark matched in beta described in Section IV. Table I lists them. The best static width is selected on the training split and then frozen for evaluation, so it represents a heuristic with hindsight over the training period but not over the test period.

C. Metrics

The two primary statistics are:

- Annualized Sharpe ratio. For each episode, computed on the equity curve at the decision cadence and annualized with the $\sqrt{\text{periods}}$ rule at 52,596 periods per year. We report the mean and a paired bootstrap confidence interval across seeds and windows.
- CVaR at 5 %. The average of the worst 5 % of episode returns, pooled across seeds, with a bootstrap confidence interval. Because the tail is thin at 13 windows × 5 seeds, we also report the worst window and its decomposition.

Secondary statistics are raw PnL, excess-versus-HODL, annualized return, Sortino ratio, maximum drawdown, rebalance count, and the cost ledger (fee income, gas,

slippage, realized impermanent loss). Every performance table reports both layers, and tables for each regime stratify by the dominant regime label of the window (bull, bear, sideways, high volatility). All comparisons against a benchmark use paired differences within each window, so market moves common to both policies do not inflate the apparent effect.

D. Ablations

The failure modes of C2 are ablated one at a time: band geometry removed from the observation; training windows held fixed instead of resampled; the hedge disabled; the reward's risk penalty toggled ($\lambda = 0$ versus $\lambda > 0$); regime features removed. The C1 economic decomposition is ablated over frictions (full model, no gas, static fee, no impermanent loss) and repeated on the 0.30 % pool to test the prediction that a higher fee tier buys a longer viable range. For C4 the runner sweeps the environment along the axes that the economics identifies: realized volatility, volume relative to liquidity, fee tier, gas price and capital. Each sweep point is evaluated on replayed windows, so the boundary is measured against real market states rather than a fitted process.

E. Hypotheses and decision rules

- C1 (limits). The decomposition of the best passive benchmark is negative in excess-versus-HODL terms and dominated by adverse selection; we report where the sign flips across regimes and fee tiers.
- C2 (failure modes). Each ablated fix degrades the learned policy relative to the full interface, or reproduces on at least one seed the degenerate behavior documented in ADR-030, in which the policy repeats one action. The claim is descriptive: we report the failure, the diagnosis and the effect of the fix.
- C3 (joint policy). The learned policy's mean Sharpe exceeds the best static width's and the hedged HODL benchmark's, with a paired 95 % confidence interval excluding zero; its 5 % CVaR is no worse than the best heuristic's; and its action distribution is not degenerate across seeds.
- C4 (boundary). We report the fitted boundary of $\text{Sharpe}_{\text{learned}} - \text{Sharpe}_{\text{passive}} = 0$ along each environmental axis, with the analytical LVR threshold from Eq. (5) overlaid as the theoretical comparison.

Rules stated in advance are the point: they prevent the result from being chosen after the fact. Where the evidence turns out negative, the negative result is reported as the finding.

F. Reproducibility

Every training run writes a manifest with its config hash, git commit, seed and library versions; every evaluation writes each episode's equity curve, cost ledger and decision log. Determinism is enforced at the level of individual components (seeded generators everywhere, no clock time in state) and exercised by the parity and look-ahead suites. Running a reported configuration again is expected to reproduce its learning curve, and a reported number without a manifest behind it is treated as a bug.

VII. Results

[PENDING: This section is intentionally a stub while the RL experiments across seeds run. It will report C1–C4 in order, and every number will come from the locked runs (manifests under results/), never from a draft memory. Nothing in this section may be filled until the runs complete.]

A. C1: Where fees fail to cover LVR and rebalancing

[PENDING: Report the passive benchmark cost decomposition (fees, impermanent loss, LVR, gas, slippage) on both fee tiers, the cost of greedy versus lazy rebalancing, and the sign of excess-versus-HODL across regimes. The preliminary diagnostic over 15 windows recorded in ADR-030 motivates the claim; the final table replaces it.]

B. C2: Failure modes in naive RL

[PENDING: Report the collapse before the fixes (policies that repeat one action, flat learning curves) and the ablations of each fix (band geometry off, fixed training window, reward measured against the rolling basket, execution at decision time off), with the evidence for each diagnosis.]

C. C3: The learned joint policy

[PENDING: Report Sharpe and CVaR out of sample for the learned policy against the full benchmark ladder of Table I, paired within windows and across seeds, plus the action distribution that shows the policy is not degenerate and its decision boundaries recover the band exit and the trade-off between fees and gas.]

D. C4: The active/passive boundary

[PENDING: Report the swept boundary where the learned and passive Sharpe ratios are equal, as a function of volatility, volume relative to liquidity, fee tier, gas and capital, with the analytical LVR threshold from Eq. (5) overlaid.]

VIII. Discussion and Threats to Validity

[PENDING: The interpretive discussion is deferred until the locked runs land: it will connect the C1–C4 results to the active/passive boundary and to the design principles for RL that accounts for friction in this setting. The validity concerns below are already fixed and do not depend on the results.]

A. What the simulator does and does not capture

Undertow replays recorded on-chain events, so market prices, liquidity and fee growth are real rather than modeled. The simulator's assumption that the agent is marginal, however, means the agent's own liquidity never moves the price, and its fee share is proportional to active liquidity. This is the standard replay assumption, and every run manifest records it, but it cannot capture queueing at the tick level or the price impact an unusually large position would have. The gas model charges representative gas units for each action plus a flat tip surcharge rather than receipts for each transaction, so gas estimates are a defensible floor, not an exact quote. The hedge leg uses a perpetual proxy margined in USDT rather than whatever venue an LP would actually use, so its funding and basis are representative rather than exact for a venue. Each of these is a modeling choice with a stated direction, not a hidden assumption.

B. Threats to validity

- Scope of one pair. The study covers one asset pair across two fee tiers. The C4 boundary is measured within that scope; extrapolating it to other pairs requires running the sweep again.
- Thin risk tail. CVaR at 5 % over thirteen windows and five seeds averages a handful of episodes. We therefore report bootstrap intervals and the worst window alongside it, and we do not treat a small CVaR difference as decisive on its own.
- Regime labels come from thresholds fixed in advance. They are a labelling device, not a discovered latent state. Results are stratified by them for interpretability, and the process that calibrates transitions is anchored to the same labels, which keeps train and evaluation consistent but makes the labels carry weight.
- Multiple comparisons. C4 sweeps several axes and C3 compares many benchmarks; paired confidence intervals and a primary metric registered in advance (Sharpe) are the mitigation, and any secondary win is labelled as such.
- The gap from simulation to reality. The parity harness bounds the gap between simulator and backtester, not the gap between backtester and reality. The latter is quantified where the data permits (fee and collect reconciliation) and discussed where it does not (latency, MEV, private order flow).

C. What we do and do not claim

We claim results about performance on recorded market states, measured by Sharpe and CVaR under a stated cost model. We do not claim deployable profitability, optimality of the action grid, or that the learned policy would survive market conditions absent from 2022–2024. Negative results, where active management does not pay, are reported as findings of C4 rather than hidden.

IX. Conclusion

[PENDING: The conclusion will be written once C1–C4 are measured. Its shape is fixed: restate the four claims with their verdicts, state the active/passive boundary as measured, and list what the study does not settle.]

Concentrated liquidity provision is a problem of impulse control under adversarial order flow, and the report's central question is whether a risk-sensitive learned policy can jointly manage band timing, band width and directional risk better than the best rules tuned by hand once gas, slippage, fees and funding are charged. To make that question answerable rather than rhetorical, this report has described Undertow: an exact on-chain replay backtester, a fast simulator checked for parity against it, a risk-sensitive formulation of the LP problem, and a protocol that reports Sharpe and CVaR alongside raw PnL. The four claims are ordered so that the negative result comes first, because a policy can only be judged against an honest baseline: C1 bounds the economics, C2 explains why naive learning fails there, and C3–C4 ask what a policy that survives can add.

Future work follows directly from the boundaries of the study: extending the joint action space with continuous width and optimal exit, running the boundary sweep across more pairs and fee tiers, modeling latency and private order flow explicitly, and replacing the representative perpetual proxy with a model of funding and basis for a specific venue. Each extension runs the same testbed again rather than redesigning it.

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